# Application of Logistic Regression Models

STAT 437 LEC A1 - Fall 2023

APPLIED STATISTICAL METHODS

Prepared by

Group-4

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# 1 Introduction

The purpose of this project was to investigate the effects of 8 variables (CAT, AGE, CHL, SMK, ECG, DBP, SBP and HPT) on coronary heart disease (CDH) using a sample of 609 observers. The selection of models and hypothesis tests will be contingent upon the type of the variables under examination. In this case, a logistic regression model will be selected and compared with the usual linear regression model. It is crucial to acknowledge that when using the logistic regression model, one must consider the over-dispersion due to the possible relationship between mean and variance. Then, two models, the full model and the restricted model will be compared to choose the final model.

# 2 Body

## 2.1 Data sets

The dataset “Evans” collected in the r-package “lbreg” is a collection of 609 observations on nine variables, which are “CDH, CAT, AGE, CHL, SMK, ECG, DBP, SBP and HPT”. Among all the nine variables, CDH is a  binary response, and the summary of each variable is labeled below in Table 1.

Table 1

| **VARIABLES** | **SUMMARY** |
|---|---|
| CDH | outcome variable; 1 = coronary heart disease |
| CAT | 1 = high, 0 = normal catecholamine level |
| AGE | age (in years) |
| CHL | cholesterol, mg/dl |
| SMK | 1 = subject has ever smoked |

| | |
|---|---|
| ECG | 1 = presence of electrocardiogram abnormality |
| DBP | diastolic blood pressure, mmHg |
| SBP | systolic blood pressure, mmHg |
| HPT | 1 = SBP greater than or equal to 160 or DBP greater than or equal to 95 |

## 2.2 Model

The logistic regression model is employed in scenarios where a binary variable, "y," exhibits only two possible values, such as 0 or 1, as observed in this project (specifically, for "CDH" in this case). It is a frequently used model in various domains, including but not limited to survival analysis (death/alive), medical diagnoses (cancer/health), and outcome prediction (success/failure). In such contexts, it is reasonable to assume that the binary response variable "y" conforms to a binomial or Bernoulli distribution. The logistic link is commonly chosen as it is the most popular one. And the link function is: $g(\mu) = log(\frac{\mu}{1-\mu})$

where $\mu = E(y) = P(y = 1)$ and the resulting GLM is:

$$log\left(\frac{\mu_i}{1-\mu_i}\right) = x_{i1}\beta_1 + x_{i2}\beta_2 + \ldots + x_{ip}\beta_p, i = 1, 2, \ldots, n$$

where $u_i = E(y_i) = P(y_i = 1)\ and\ y_i = 1$ is assumed to follow a Binomial distribution with mean $u_i$.

In this project, residuals are not well-defined for logistic regression models since the response variable's value is either 0 or 1, which means that residual plots will not be very useful for checking the goodness-of-fit. Therefore, using binary logistic regression to estimate the probability of coronary heart disease given by the other eight variables.

## 2.3 Output

Firstly, install and library the package "faraway", "lbreg" and "MASS" needed in this project. Then import the dataset "Evans".

**Table for the first 6 data**:

| | CDH <int> | CAT <int> | AGE <int> | CHL <int> | SMK <int> | ECG <int> | DBP <int> | SBP <int> | HPT <int> |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 56 | 270 | 0 | 0 | 80 | 138 | 0 |
| 2 | 0 | 0 | 43 | 159 | 1 | 0 | 74 | 128 | 0 |
| 3 | 1 | 1 | 56 | 201 | 1 | 1 | 112 | 164 | 1 |
| 4 | 0 | 1 | 64 | 179 | 1 | 0 | 100 | 200 | 1 |
| 5 | 0 | 0 | 49 | 243 | 1 | 0 | 82 | 145 | 0 |
| 6 | 0 | 0 | 46 | 252 | 1 | 0 | 88 | 142 | 0 |

6 rows

Next, fit binary logistic regression as our comprehensive model. Given that the variable "CHD" is binary (0 and 1), it necessitates logistic regression, while the usual linear regression model is inappropriate for this analysis. Consequently, we execute logistic regression using all the predictors, namely CAT, AGE, CHL, SMK, ECG, DBP, SBP, and HPT. The summary of fitted model is illustrated by the following table:

**Summary for fit1:**

```
Call:
glm(formula = CDH ~ ., family = binomial, data = Evans)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.764709   1.579763  -4.282 1.85e-05 ***
CAT          0.750257   0.377231   1.989  0.04672 *
AGE          0.037781   0.016031   2.357  0.01844 *
CHL          0.008674   0.003296   2.632  0.00850 **
SMK          0.867474   0.307218   2.824  0.00475 **
ECG          0.403426   0.296125   1.362  0.17309
DBP          0.012402   0.014284   0.868  0.38526
SBP         -0.010571   0.008073  -1.309  0.19038
HPT          0.532752   0.383747   1.388  0.16505
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 438.56  on 608  degrees of freedom
Residual deviance: 398.60  on 600  degrees of freedom
AIC: 416.6
```

```
Number of Fisher Scoring iterations: 5
```

For checking the over-dispersion problem in our data, the quasibinomial model is used for getting the dispersion parameter. If the dispersion parameter equals or is close to 1, the response variable follows a binomial distribution.

**Summary of checking the over-dispersion:**

```
Call:
glm(formula = CDH ~ ., family = quasibinomial, data = Evans)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -6.764709   1.581990  -4.276 2.21e-05 ***
CAT          0.750257   0.377763   1.986  0.04748 *
AGE          0.037781   0.016054   2.353  0.01892 *
CHL          0.008674   0.003301   2.628  0.00881 **
SMK          0.867474   0.307651   2.820  0.00497 **
ECG          0.403426   0.296543   1.360  0.17420
DBP          0.012402   0.014304   0.867  0.38628
SBP         -0.010571   0.008084  -1.308  0.19150
HPT          0.532752   0.384288   1.386  0.16616
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for quasibinomial family taken to be 1.002821)

    Null deviance: 438.56  on 608  degrees of freedom
Residual deviance: 398.60  on 600  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

The results reveal an estimated dispersion parameter 1.002821, indicating a proximity to 1. This observation suggests that the assumed binomial distribution is plausible, proving over-dispersion problem does not exist. Consequently, these findings bolster the reliability of the results obtained through the logistic regression model.

For finding significant co-variate, the stepwise method is used by the stepAIC() command in R. The final model with the smallest AIC is given below:

```
Step:  AIC=413.95
CDH ~ CAT + AGE + CHL + SMK + HPT
```

Summary of the restricted test:

**Summary for fit2:**

……

```
Call:
glm(formula = CDH ~ CAT + AGE + CHL + SMK + HPT, family = binomial,
    data = Evans)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.680112   1.136363  -5.879 4.14e-09 ***
CAT          0.715810   0.340180   2.104  0.03536 *
AGE          0.032770   0.015197   2.156  0.03105 *
CHL          0.008608   0.003259   2.641  0.00827 **
SMK          0.802906   0.303001   2.650  0.00805 **
HPT          0.476272   0.289296   1.646  0.09970 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 438.56  on 608  degrees of freedom
Residual deviance: 401.95  on 603  degrees of freedom
AIC: 413.95

Number of Fisher Scoring iterations: 5
```

The stepwise method based on AIC values removes the “ECG”, “DBP”, and “SBP” variables, which shows that the other five variables influence CHD more significantly.

Compare the two nested models by using a deviance test:

**ANOVA test for fit2 and fit1:**

```
Analysis of Deviance Table

Model 1: CDH ~ CAT + AGE + CHL + SMK + HPT
Model 2: CDH ~ CAT + AGE + CHL + SMK + ECG + DBP + SBP + HPT
  Resid. Df Resid. Dev Df Deviance
1       603     401.95
2       600     398.60  3   3.3468
```

The analysis of results reveals a difference in deviances of 3.3468 between the two models, with 3 degrees of freedom. These findings indicate that models 1 and 2 do not differ significantly. The associated p-value of 0.340744 leads to the selection of the smaller model with fewer predictors, which is "fit2."

In this case the Wald-type z-test is not used because the model has several predictors and they are not continuous.

# 3 Conclusion

The final GLM model selected GLM is:

$$log\left(\frac{\mu_i}{1-\mu_i}\right) = x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i3}\beta_3 + x_{i4}\beta_4 + x_{i5}\beta_5, x_{i1} = CAT, x_{i2} = AGE, x_{i3} = CHL, x_{i4} = SMK$$

$$x_{i5} = HPT$$

For multiple noncontinuous predictors of the data set, the Wald-type test was not selected. However, the restricted model and the full model are similar. Therefore the restricted model is preferred.

The usual linear regression model was unsuitable for this analysis due to the binary response variable, CHD. Furthermore, the over-dispersion parameter derived from the quasibinomial model showed it to be very close to 1, suggesting that over-dispersion is not a concern.

Significant covariates identified in the analysis are CAT, AGE, CHL, SMK under $\alpha$=0.05.

# 4 Appendix

## 4.1 Contribution

Chenyu Pan: Code and report (introduction and conclusion)

Enzhe Wang: Report (Body and Appendix)

Luoxi Han: Slides and presentation

## 4.2 Reference

1. Chapter-2 PDF in eclass
https://eclass.srv.ualberta.ca/pluginfile.php/10382956/mod_resource/content/1/Chapter-2%28Final%20GLM%29.pdf

2. R-package “lbreg”
https://cran.r-project.org/web/packages/lbreg/lbreg.pdf

3. R-package “faraway”
https://cran.r-project.org/web/packages/faraway/faraway.pdf

4. R-package “MASS”
https://cran.r-project.org/web/packages/MASS/MASS.pdf

## 4.3 R-code

```
rm(list=ls())
library(faraway)
library(lbreg)
library(MASS)
data(Evans)

# Fit the logistic regression with all predictors CAT, AGE, CHL, SMK, ECG, DBP, SBP,and HPT.
## Fit binary logistic regression as a Full model ##
fit1 = glm(CDH ~ .,family = binomial, data = Evans)
summary(fit1)
```

```
## Stepwise method for variable selection ##
fit2= stepAIC(fit1,direction = c("both"),k = 2)
summary(fit2)

################### Compare the two nested models using deviance test####
anova(fit2, fit1)
# the difference in deviances from the two models is 3.35
# the two models differ by 3 parameters (3 degrees of freedom).
# we see that models 1 and 2 are not significantly different
# (at 5 percentage level), and the p-value of the 0.340744
# choose the smaller model(Restricted model or fitted model), which is fit2 (CDH ~ CAT + AGE + CHL
+ SMK + HPT)

################### To check over-dispersion ###
fit3 = glm(CDH ~ .,family = quasibinomial, data = Evans)
summary(fit3)
# The estimated dispersion parameter is 1.002821, very close to 1. Thus, the assumed
# binomial distribution may hold or there may be no over-dispersion problem, suggesting
# that the results from fit2 may be reliable.
```